

An Exploration of Washington DC., Tree Phenology Drivers

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Introduction

Phenological processes, such as the timing of leaf-out and senescence, are highly sensitive to environmental conditions and serve as key indicators of ecosystem response to climate change. In urban environments, these processes are further influenced by anthropogenic factors such as heat island effects and land cover modification.

This study examines the relationship between climate variables and tree phenology, with particular attention to species-level differences. Specifically, we asks:

How do temperature, precipitation, and urban environmental factors influence phenological timing, and how do these effects vary across species?

Rather than focusing on prediction, this study adopts an explanatory framework, aiming to quantify the magnitude of environmental influences and identify key drivers of phenological variation.

Literature Review

Urban environments, which experience elevated temperatures relative to surrounding areas, provide a useful proxy for understanding how future climate warming may influence forest phenology. Because cities are typically warmer, they are often used to examine how rising temperatures shift the timing of key phenological events such as the start and end of the growing season. However, phenological patterns in urban areas are not driven by temperature alone. Additional factors—including species composition, plant functional type, and local site characteristics such as impervious surface cover—can also influence observed patterns, making these effects difficult to disentangle without high-resolution data. Using species-level observations and satellite imagery, Michael Alonzo et al. (2023) find that species identity explains substantially more variation in phenological timing than local environmental conditions,

suggesting that species composition plays a dominant role in shaping urban phenological responses.

More broadly, tree phenology—the timing of key life cycle events such as leaf emergence and senescence—is a critical adaptive trait shaped by both environmental conditions and internal developmental processes. These processes are regulated by factors such as temperature and moisture and are closely linked to biological mechanisms like meristem activity and phase change. Phenological responses vary across species, reflecting differences in ecological strategies and sensitivity to environmental stressors (Brunner et al., 2017). This species-level variation is particularly relevant in urban environments, where heterogeneous species composition can strongly influence aggregate phenological patterns. For example, Zhiwei Wu et al. (2025) show that differences in phenological responses between urban and rural areas are driven more by variation in species composition than by temperature alone, highlighting the importance of accounting for differences across species in empirical analyses.

Phenology is also widely used as a bioindicator of climate change due to its strong statistical relationship with temperature. In urban contexts, this relationship is often amplified by the urban heat island effect, allowing cities to serve as a proxy for future climatic conditions. However, this relationship may be weakened by additional environmental and site-specific factors, complicating interpretation. Susanne Jochner and Annette Menzel (2015) emphasize that variation in study design, environmental conditions, and local characteristics can influence observed phenological patterns. These findings underscore the importance of incorporating multiple explanatory variables—such as temperature, precipitation, land cover, and species identity—when modeling phenological responses in urban environments.

Recent advances in phenological modeling have increasingly focused on improving predictive performance under changing climate conditions. Machine learning (ML) and deep learning approaches are able to capture complex, non-linear relationships between environmental drivers and phenological responses and have demonstrated strong predictive accuracy. For instance, Vivien Sainte Fare Garnot et al. (2026) show that deep learning models can outperform traditional statistical approaches and perform comparably to process-based models, particularly under shifting climatic conditions. Similarly, Chengxi Gao et al. (2023) find that machine learning approaches improve prediction accuracy and better capture interactions among multiple environmental and geographic variables. However, while these approaches are well-suited for prediction, they often prioritize accuracy over interpretability.

In contrast, linear regression models provide a transparent and interpretable framework for isolating and quantifying the effects of individual environmental variables on phenological timing. By controlling for multiple factors simultaneously, this approach allows for direct estimation of the influence of temperature, precipitation, and species identity, making it particularly well-suited for explanatory analyses of phenological variation. While machine learning approaches may improve predictive performance, linear regression enables clearer inference on the drivers of phenological change, which is central to understanding how environmental factors shape phenology across species and urban conditions.

Urban environments, particularly heterogeneous cities like Washington, DC, provide a valuable setting for examining how climate and land cover interact to shape tree phenology. The availability of high-resolution remote sensing data, combined with detailed measures of urban form such as impervious surface cover, allows for fine-scale analysis of both environmental conditions and species-level variation. Building on prior work highlighting the roles of climate, species composition, and urban conditions in shaping phenology, this study employs an explanatory modeling framework to quantify these relationships. By estimating a series of linear regression models that relate phenological metrics to climate variables, land cover characteristics, and species identity, this approach allows us to isolate the marginal effects of each predictor while controlling for confounding factors, providing insight into the drivers of phenological variation across species and environments.

Methodology

This study integrates remotely sensed phenological data with climate and land cover information to examine the drivers of urban tree phenology. Phenological metrics—including start of season (SOS), end of season (EOS), and transitional measures such as spring green-up and autumn senescence rates—were derived from high-resolution satellite imagery, providing repeat observations of individual trees across multiple years. These data were matched with climate variables obtained from NOAA, including seasonal averages of temperature and precipitation, which capture key environmental conditions influencing plant development. To account for urban heterogeneity, land cover characteristics were incorporated using measures of impervious surface cover within a defined spatial buffer around each tree, representing localized urban intensity. In addition, species identity was included as a categorical variable to capture interspecific variation in phenological responses. Together, these variables allow for a comprehensive assessment of how climate, urban form, and species composition jointly influence phenological timing at the individual tree level.

Data Sourcing:

This study integrates remotely sensed phenological data, climate observations, and land cover information to construct a tree-level panel dataset for Washington, DC. Phenological metrics were derived from high-resolution satellite imagery obtained from PlanetScope, which provides near-daily multispectral imagery at fine spatial resolution. These data enable repeated observations of individual trees across time and allow for precise estimation of seasonal vegetation dynamics.

Climate data were obtained from the National Oceanic and Atmospheric Administration (NOAA) Climate Data Online (CDO) database. Daily observations of temperature and precipitation were collected from weather stations in and around the study area. These data were used to construct seasonal climate variables relevant to plant development.

To capture urban environmental heterogeneity, Impervious surface data were provided by Alonzo et al. and linked to individual tree locations, while additional land cover variables were

obtained from the National Land Cover Database (NLCD) and extracted at the same spatial scale.

Remote Sensing & Phenological Metric Construction:

Satellite imagery was first filtered by spatial extent and time period to match the study area and observation window. Images were screened for quality, including cloud cover and atmospheric interference, and processed to compute vegetation indices, specifically the Normalized Difference Vegetation Index (NDVI). NDVI time series were then extracted at individual tree locations.

Phenological metrics were derived from NDVI time series using a curve-fitting and smoothing approach implemented through phenology modeling code originally developed by Michael Alonzo. The code was adapted for this study to align with the Washington, DC dataset, including adjustments to temporal coverage, variable naming conventions, and data structure. These modifications ensured compatibility with the study dataset while preserving the underlying modeling framework.

The resulting outputs include key phenological metrics such as the start of season (SOS), end of season (EOS), and intermediate transition measures (e.g., spring green-up and autumn senescence rates). These variables serve as observable proxies for underlying biological processes such as meristem activity and seasonal growth dynamics.

Figure 1: Washington DC. Tree Spatial Distribution Observation

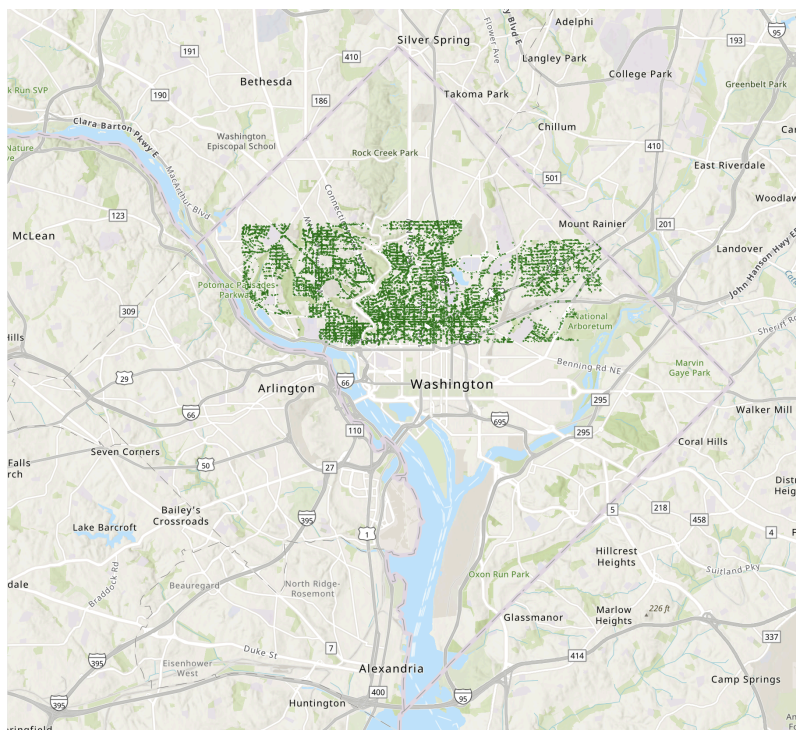


Figure 1: Spatial distribution of tree observations used in the analysis across Washington, DC. Points represent individual trees, highlighting clustering patterns and coverage across the urban landscape.

Species Selection:

To ensure meaningful and consistent analysis, this study focuses on a curated subset of tree species commonly found within the Washington, DC urban forest. Species were selected based on three primary criteria: data availability, temporal consistency, and ecological relevance.

First, species were required to have sufficient observations across multiple years to support reliable estimation of phenological metrics. This ensured that each species included in the analysis had adequate temporal coverage for both the remote sensing data and associated climate variables. Species with sparse observations or inconsistent temporal records were excluded to avoid introducing noise or bias into the results.

Second, the selected species represent a mix of native and non-native trees, as well as a range of genera and functional types. This diversity allows the analysis to capture variation in phenological responses associated with differences in biological traits, ecological strategies, and adaptation to urban conditions. Including both native and widely planted ornamental species is particularly important in urban environments, where species composition is shaped by human management.

Finally, species were chosen based on their prevalence within the study area, ensuring that results reflect common components of the urban forest rather than rare or atypical cases. This improves the generalizability of the findings to broader urban ecological contexts.

Together, these criteria result in a dataset that balances biological diversity with statistical reliability, enabling robust analysis of how climate and urban factors influence phenological timing across species.

Table 1: Tree Species Included in Analysis

SpCode	Latin Name	Genus	Origin	Native Status	Common Name
ACPL	<i>Acer platanoides</i>	Acer	Europe	Non-native	Norway Maple
ACRU	<i>Acer rubrum</i>	Acer	USA-East	Native	Red Maple
ACSA2	<i>Acer saccharum</i>	Acer	USA-East	Native	Sugar Maple
CECA	<i>Cercis canadensis</i>	Cercis	USA-East	Native	Eastern Redbud

CEOC	<i>Celtis occidentalis</i>	Celtis	USA	Native	Common Hackberry
GIBI	<i>Ginkgo biloba (female)</i>	Ginkgo	Asia-South	Non-native	Ginkgo
GLTR	<i>Gleditsia triacanthos var. inermis</i>	Gleditsia	USA	Native	Thornless Honeylocust
GYDI	<i>Gymnocladus dioicus</i>	Gymnocladus	USA-NE	Native	Kentucky Coffee Tree
LAIN	<i>Lagerstroemia indica</i>	Lagerstroemia	Asia	Non-native	Crapemyrtle
LIST	<i>Liquidambar styraciflua 'Sterile'</i>	Liquidambar	USA-East	Native	Sweetgum
LITU	<i>Liriodendron tulipifera</i>	Liriodendron	USA-East	Native	Tulip Tree
NYSY	<i>Nyssa sylvatica</i>	Nyssa	USA-East	Native	Blackgum
PLAC1	<i>Platanus × acerifolia</i>	Platanus	Europe	Non-native	London Plane
PLOC	<i>Platanus occidentalis</i>	Platanus	USA-SE	Native	American Sycamore
PRKW	<i>Prunus serrulata 'Kwanzan'</i>	Prunus	Asia	Non-native	Cherry Blossom
PRYE	<i>Prunus yedoensis</i>	Prunus	Asia	Non-native	Yoshino Cherry
QUAC	<i>Quercus acutissima</i>	Quercus	Asia	Non-native	Sawtooth Oak
QUBI	<i>Quercus bicolor</i>	Quercus	USA-NE	Native	Swamp White Oak
QUCO	<i>Quercus coccinea</i>	Quercus	USA-NE	Native	Scarlet Oak
QULY	<i>Quercus lyrata</i>	Quercus	USA-SE	Native	Overcup Oak
QUPA	<i>Quercus palustris</i>	Quercus	USA-NE	Native	Pin Oak
QUPH	<i>Quercus phellos</i>	Quercus	USA-SE	Native	Willow Oak
QURU	<i>Quercus rubra</i>	Quercus	USA-East	Native	Red Oak

TIAM	<i>Tilia americana</i>	Tilia	USA-East	Native	American Linden
TICO	<i>Tilia cordata</i>	Tilia	Europe	Non-native	Little Leaf Linden
TITO	<i>Tilia tomentosa</i>	Tilia	Europe	Non-native	Silver Linden
ULAM	<i>Ulmus americana 'Jefferson'</i>	Ulmus	USA-East	Native	American Elm
ULPA	<i>Ulmus parvifolia</i>	Ulmus	Asia	Non-native	Chinese Elm
ZESE	<i>Zelkova serrata 'Green Vase'</i>	Zelkova	Asia	Non-native	Japanese Zelkova

Table 1: Tree species included in the analysis, spanning multiple genera, geographic origins, and native statuses within the Washington, DC urban forest.

Climate, Land Cover, & Urban Covariates:

Daily climate observations from NOAA were cleaned and aggregated into seasonal variables aligned with biologically relevant periods. Temperature variables were constructed as seasonal averages (e.g., spring mean temperature), while precipitation was summarized using total accumulation and frequency measures (e.g., precipitation days). These aggregated variables capture the environmental conditions most relevant for phenological development.

Urban conditions were measured using a continuous indicator of impervious surface cover derived from high-resolution spatial data, representing the proportion of impervious area within a 90-meter buffer surrounding each tree. This variable provides a precise, tree-level measure of urban intensity and allows for the estimation of marginal effects of urbanization on phenological timing.

In addition, categorical land cover variables from the National Land Cover Database (NLCD) are incorporated in alternative model specifications. These variables classify each tree's surrounding environment into discrete land cover types (e.g., developed high intensity, open space). While these categories provide useful contextual information, they are coarser than the continuous impervious surface measure and are therefore used primarily as robustness checks rather than the primary measure of urban conditions.

Washington DC. Dataset Construction:

All datasets were merged at the individual tree level to create a panel dataset containing phenological metrics, climate variables, land cover characteristics, and species identity. Temporal alignment was performed to ensure consistency between phenological observations

and climate variables within each growing season. Additional data cleaning steps included the removal of duplicates, handling of missing values, and consistency checks across years.

Regression Analysis:

To examine the drivers of phenological variation, this study employs an explanatory modeling framework using linear regression applied to a tree-level panel dataset. The dataset consists of repeated observations of individual trees across multiple years, allowing for both cross-sectional variation and temporal variation. The primary objective is to quantify how climate variables, urban conditions, and species identity are associated with variation in phenological timing.

We estimate a series of pooled ordinary least squares (OLS) models of the form:

Baseline Model:

$$Phenology_{i,t} = \beta_0 + \beta_1 Temperature_t + \beta_2 Precipitation_t + \beta_3 Impervious_i + \beta_4 DBH_i + \gamma_s + \epsilon_{i,t}$$

Where $Phenology_{i,t}$ represents a phenological outcome for tree i in year t , including the start of season (SOS), summer shoulder, autumn shoulder, and end of season (EOS). Temperature and precipitation vary at the yearly level and capture interannual climate conditions, while impervious surface and DBH vary at the tree level and capture spatial and structural differences across individuals. Species is included as a set of indicator variables (γ_s) to account for biological differences across tree types.

For each phenological outcome, I estimated a linear regression model using spring temperature, spring precipitation, DBH, and impervious surface as predictors, while including species fixed effects to account for differences across tree species. We use spring temperature across all models because it's the strongest and most consistent driver of phenology, and early-season conditions often influence the timing of the entire growing season. However, we acknowledge that using only spring climate variables may not fully capture late-season dynamics, which is why autumn models tend to have lower explanatory power.

Phenology-Specific Models

$$Y_{i,t} = \beta_0 + \beta_1 spring_avg_t + \beta_2 spring_prcp_t + \beta_3 impervious_{90m,i} + \beta_4 DBH_i + \gamma_s + \epsilon_{i,t}$$

Where, $Y_{i,t} \in \{SOS, SummerShoulder, AutumnShoulder, EOS\}$

Spring temperature is used across all models because it represents the strongest and most consistent driver of vegetation dynamics. Early-season conditions often influence the timing of the entire growing season, including later phenological transitions.

However, this specification has limitations. Because only spring climate variables are included, the model may not fully capture late-season processes such as senescence. This is reflected in the lower explanatory power observed in autumn-related model

This pooled specification leverages both temporal and cross-sectional variation to estimate average relationships between environmental conditions and phenological outcomes. Temporal variation in climate variables allows for identification of how interannual differences in temperature and precipitation influence phenology, while cross-sectional variation in impervious surface and species identity captures heterogeneity in local urban conditions and biological responses.

Fixed Effects Specification & Robustness Check:

To account for unobserved temporal shocks that may affect all trees within a given year, we estimate additional models including year fixed effects:

$$Phenology_{i,t} = \beta_1 Temperature_t + \beta_2 Precipitation_t + \beta_3 Impervious_i + \beta_4 DBH_i + \beta_5 Species_i + \gamma_t + \delta_t + \epsilon_{i,t}$$

where δ_t represents year-specific intercepts capturing unobserved factors that vary across years but are common to all trees, such as anomalous climate conditions, measurement differences, or other temporal influences.

While the inclusion of year fixed effects improves model fit and controls for temporal confounding, it introduces an important identification tradeoff. Because climate variables vary primarily at the yearly level, they are highly collinear with the year fixed effects. As a result, the fixed effects specification absorbs much of the variation used to estimate the effects of temperature and precipitation, leading to changes in magnitude and, in some cases, instability in climate coefficients.

Accordingly, the fixed effects models are not used as the primary specification for interpreting climate effects. Instead, they serve as robustness checks to assess whether the estimated effects of urban conditions and species identity persist after controlling for unobserved temporal variation.

Coefficient Interpretation:

Coefficients from the pooled OLS models are interpreted as the marginal effect of each explanatory variable on phenological timing, holding all other factors constant. For example, the coefficient on spring temperature represents the expected change in phenological timing (measured in days) associated with a one-degree increase in temperature. Similarly, the coefficient on impervious surface reflects the change in phenological timing associated with increased urban intensity.

Because climate variables vary at the yearly level, their coefficients capture how interannual differences in temperature and precipitation translate into shifts in phenological timing across the full sample. In contrast, coefficients on impervious surface and DBH capture cross-sectional differences across trees within the urban environment.

As the models are estimated within an explanatory framework, the focus is on the direction, magnitude, and statistical significance of relationships rather than predictive performance. This approach allows for direct inference on the relative importance of climate conditions, urban context, and species-level differences in shaping phenological responses.

Results

A series of linear regression models were estimated to examine the relationship between climate variables, urban context, and tree phenological metrics. Model performance varied across phenological stages.

Early-season models exhibited stronger explanatory power, with R^2 values of 0.151 for start of season (SOS) and 0.245 for summer transition metrics. In contrast, late-season models showed substantially lower explanatory power, with R^2 values of 0.083 for autumn transition metrics and 0.078 for end-of-season (EOS) .

Across all models, F-statistics were highly significant ($p < 0.001$), indicating that the included predictors jointly explain variation in phenological outcomes.

Climate Drivers:

Spring temperature and precipitation were statistically significant predictors across all models.

Spring temperature exhibited a strong and consistent relationship with phenological timing. A one-degree increase in spring temperature was associated with an approximately 3.50-day earlier start of season ($p < 0.001$). Similar magnitudes were observed for summer transitions (-3.82, $p < 0.001$), while autumn and EOS models showed smaller but still significant positive relationships (1.51 and 0.77, respectively; $p < 0.001$) .

Spring precipitation was also highly significant across models, though the direction of the relationship varied. Positive associations were observed for SOS (0.92, $p < 0.001$), summer (2.03, $p < 0.001$), and EOS (1.07, $p < 0.001$), while autumn transitions exhibited a negative relationship (-1.18, $p < 0.001$) .

Urban Context & Additional Covariates:

Urban conditions, measured through impervious surface cover, were statistically significant across all models.

Impervious surface was positively associated with SOS (3.91, $p < 0.001$), autumn transitions (15.06, $p < 0.001$), and EOS (3.68, $p < 0.001$), while showing a negative relationship with summer transition metrics (-3.60, $p < 0.001$). The magnitude of the coefficient in the autumn model was notably larger than in other models.

Tree size (DBH) showed mixed results across models. It was not statistically significant in the SOS model but was significant in summer, autumn, and EOS models, with varying directions of association.

Species indicator variables were jointly significant across models, indicating meaningful differences in phenological timing across species.

Table 2: Baseline Regression Results (Pooled OLS)

Variable	SOS	Summer Shoulder	Autumn Shoulder	EOS
Spring Temperature	-3.50***	-3.82***	1.51***	0.77***
Spring Precipitation	0.92***	2.03***	-1.18***	1.07***
Impervious Surface	3.91***	-3.60***	15.06***	3.68***
DBH	0.0(n.s.)	0.024***	0.035***	-0.112***
Species Controls	Yes	Yes	Yes	Yes
Observations	82,187	82,187	82,187	82,187
R ²	0.151	0.245	0.083	0.078
F-statistic	457.20***	832.27***	231.99***	216.46***

Notes: Standard errors omitted for brevity. *** $p < 0.001$. “n.s.” indicates not statistically significant.

Fixed Effects Models

To account for unobserved temporal variation, models including year fixed effects were estimated.

The inclusion of fixed effects substantially improved model fit. The R² for the SOS model increased from 0.151 to 0.388, while summer increased from 0.245 to 0.357, autumn from 0.083 to 0.135, and EOS from 0.078 to 0.194.

Despite these improvements in fit, the magnitude and statistical significance of key predictors—particularly spring temperature and precipitation—remained consistent across specifications.

Table 3: Fixed Effects Regression Results (Year Fixed Effects)

Variable	SOS	Summer Shoulder	Autumn Shoulder	EOS
Spring Temperature	-3.01**	-6.79***	-3.59**	-7.54***
Spring Precipitation	7.25***	2.46***	-7.80***	-4.88***
Impervious Surface	4.92***	-3.08***	18.40***	3.40***
DBH	0.001 (n.s.)	0.023***	0.025***	-0.115***
Species FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	82,187	82,187	82,187	82,187
R ²	0.388	0.357	0.135	0.194

Notes: *** p < 0.001. Models include both species and year fixed effects.

Analysis & Discussion

This study examines how climate conditions, urban form, and species identity shape tree phenology in an urban environment using high-resolution, tree-level data. The results demonstrate that phenological responses are driven by a combination of climatic and non-climatic factors, with their relative importance varying across stages of the growing season. In particular, the findings highlight strong climate sensitivity in spring phenology, persistent and robust effects of urbanization, and substantial variation across species.

Climate Controls:

Across all models, temperature and precipitation emerge as key drivers of early-season phenology. Warmer spring temperatures are consistently associated with earlier start of season (SOS) and faster progression into summer, reflecting the role of temperature in regulating physiological processes such as budburst and leaf expansion (Brunner et al., 2017). Precipitation is also statistically significant across models, though its effects vary by season, suggesting a more complex influence on early growth conditions.

The influence of climate variables does however, become weaker and less consistent for late-season metrics. Autumn shoulder and end-of-season (EOS) models exhibit lower explanatory power and more variable coefficients, possibly indicating that fall phenology is governed by a broader and more complex set of controls. This aligns with prior work showing that autumn processes depend not only on temperature, but also on photoperiod, accumulated stress, and species-specific strategies (Gao et al., 2023).

The comparison between pooled and fixed effects models further underscores the role of temporal variation in climate drivers. While fixed effects models substantially improve model fit, they also produce less stable and sometimes contradictory estimates of climate effects. This reflects an identification constraint: because temperature and precipitation vary primarily at the yearly level, they are highly collinear with year fixed effects. As a result, pooled models provide more reliable estimates of climate sensitivity, while fixed effects specifications are best interpreted as robustness checks rather than primary estimates.

Urban Effects and the Role of Impervious Surface:

A central contribution of this study is the identification of strong and consistent effects of urban conditions on phenology. Impervious surface, used as a continuous measure of urban intensity, is statistically significant across all models and remains robust to the inclusion of year fixed effects. This indicates that spatial variation in urban environments plays an important role in shaping phenological outcomes independent of interannual climate variability.

The direction of these effects varies across phenological stages. Higher levels of impervious surface are associated with shifts in early-season timing and accelerated summer development, suggesting that urban environments influence not only the onset of phenological events but also the rate at which they progress. In contrast, the large positive effect observed in autumn indicates delayed senescence and an extension of the growing season in more urbanized areas.

These findings complicate the common assumption that urban warming uniformly advances phenology. While cities are often treated as proxies for future climate conditions (Jochner & Menzel, 2015), the results suggest that urban phenological responses cannot be explained by temperature alone. Instead, urban environments likely introduce additional stressors—such as altered soil conditions, water availability, and management practices—that interact with climatic drivers in complex ways. This interpretation is consistent with Alonzo et al. (2023), who show that species composition and landscape heterogeneity can outweigh temperature effects in determining urban phenological patterns.

Species Composition and Biological Variation:

Species identity accounts for a substantial share of variation in phenological timing across all models, reinforcing the importance of biological differences in shaping responses to environmental conditions. Many species exhibit statistically significant deviations in timing

relative to the reference category, with differences on the order of several days—magnitudes comparable to those of key climate variables.

These results are consistent with recent work emphasizing the role of species composition in determining phenological responses to environmental change. Wu et al. (2025) demonstrate that variation across species in temperature sensitivity can exceed the effects of climate itself in urban environments. Similarly, Brunner et al. (2017) highlight the role of genetic and physiological controls in shaping phenological timing.

The persistence of strong species effects across all model specifications indicates that phenological responses cannot be fully understood without accounting for taxonomic composition. This is particularly important in urban environments, where species selection is influenced by human management. As a result, the composition of urban forests can directly shape ecosystem-level responses to environmental change.

Species-Specific Insights:

To further illustrate how species-level differences shape phenological responses, we highlight two representative species: *Ginkgo biloba* and *Ulmus americana*. These examples demonstrate how biological traits and ecological strategies influence phenological timing in ways that extend beyond climate and urban factors alone.

***Ginkgo biloba* (Ginkgo)**

Ginkgo biloba exhibits distinctive phenological behavior across the models, particularly in late-season dynamics. The results show a strong positive association with autumn transition metrics, indicating that Ginkgo tends to maintain greenness later into the season relative to the reference species. This extended growing period may reflect its known physiological resilience and tolerance to urban stressors such as pollution, soil compaction, and heat.

Ginkgo is a widely planted non-native species in urban environments due to its durability and resistance to environmental stress. Its delayed senescence suggests that certain species are better adapted to urban conditions, allowing them to extend biological activity even under suboptimal conditions. This pattern highlights how species-specific traits can interact with urban environments to shape phenological outcomes in ways that are not captured by climate variables alone.

***Ulmus americana* (American Elm)**

In contrast, *Ulmus americana* exhibits one of the strongest negative associations with late-season phenology, particularly in the autumn and EOS models. The magnitude of these coefficients suggests that American elm enters senescence significantly earlier than many other species in the dataset.

This earlier decline may reflect greater sensitivity to environmental stressors or differences in physiological strategy compared to more resilient species like Ginkgo. While American elm is native to the region, it has historically been affected by disease and urban stress (University of Kentucky Urban Forest Initiative, n.d.), which may contribute to earlier seasonal decline under urban conditions.

The contrast between Ginkgo and American elm highlights the importance of species selection in urban ecosystems. While some species are able to maintain extended periods of activity, others exhibit earlier senescence (see figure 2), leading to substantial variation in phenological timing across the urban forest.

These species-level contrasts reinforce that phenological responses are not uniform across urban forests, and that species composition plays a critical role in determining how ecosystems respond to environmental change.”

Figure 1: Species Level Phenological Timing

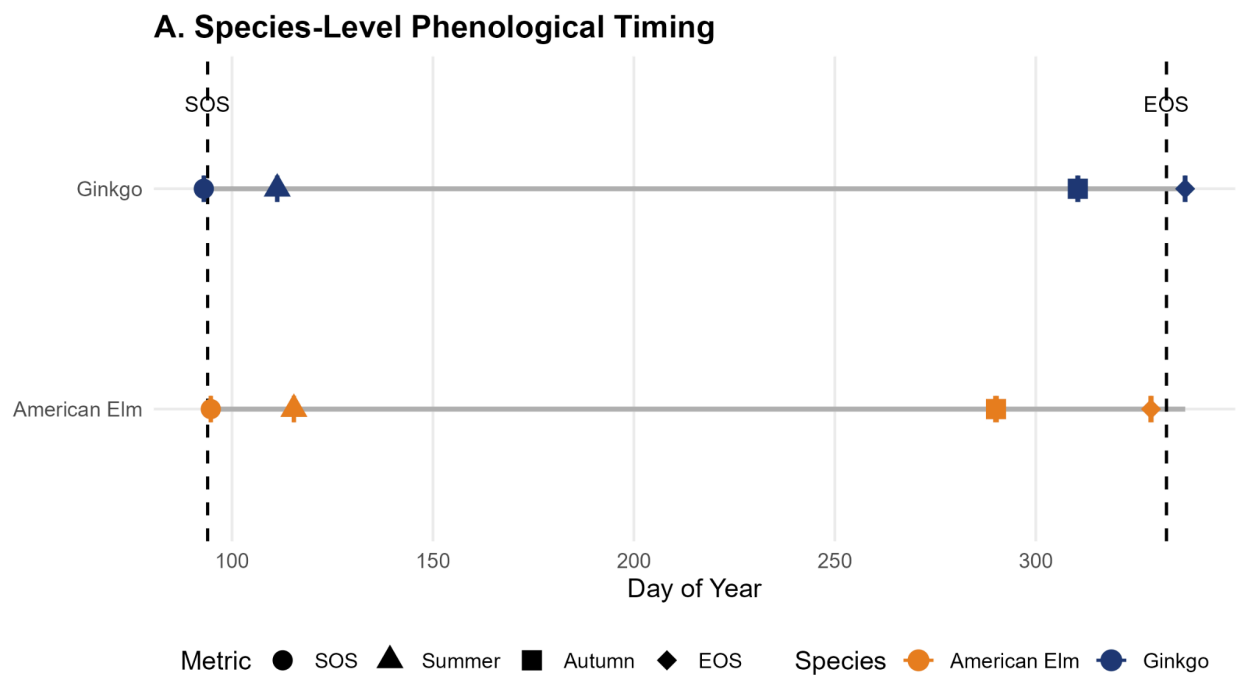


Figure 1. Species-level phenological timing for Ginkgo and American Elm. Points represent the mean day of year for key phenological metrics, with error bars showing standard errors. Dashed lines indicate average start (SOS) and end (EOS) of season across species.

Pearson Correlation Coefficient:

Pearson correlation analysis reveals clear differences in how individual species respond to environmental variables. Across species, spring temperature shows a consistent negative correlation with early-season phenological metrics, particularly start of season (SOS) and summer transitions, indicating that warmer conditions are associated with earlier development. In contrast, correlations with late-season metrics are weaker and more variable, suggesting greater heterogeneity in autumn and end-of-season dynamics. Precipitation exhibits more mixed relationships, with generally weaker correlations across species and phenological stages. Notably, the strength and direction of correlations vary across species, reinforcing the importance of species-level differences in shaping phenological responses. Overall, these results are consistent with the regression findings, showing strong and consistent climate effects in early-season phenology and more variable, less predictable relationships in later stages of the growing seasons. These patterns highlight that while climate drivers exert broad influence, the magnitude and consistency of these relationships depend on species-specific responses.

Figure 2: Pearson Correlation Coefficient

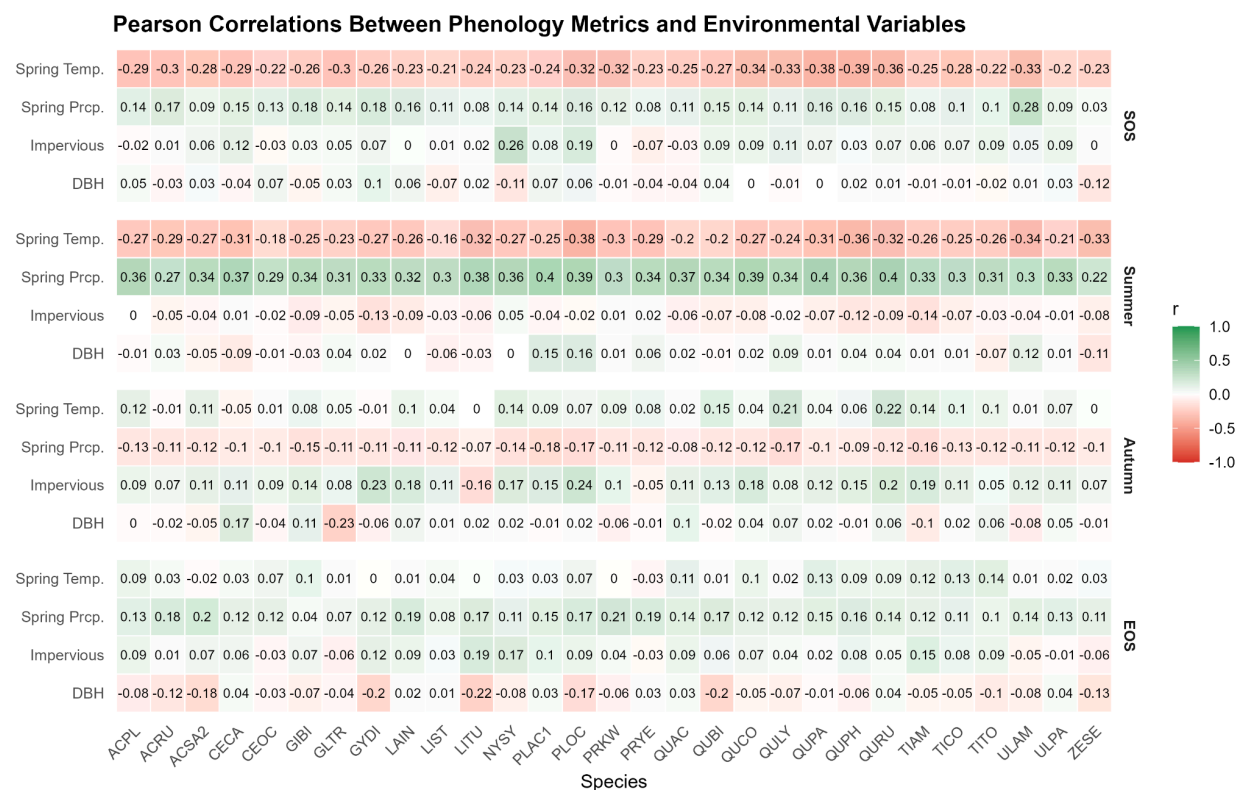


Figure 2. Pearson correlation coefficients between phenological metrics and environmental variables by species. Green cells indicate positive correlations, red cells indicate negative correlations, and darker colors represent stronger relationships.

When comparing *Ginkgo biloba* and *Ulmus americana*. For both species, spring temperature shows a negative correlation with early-season phenological metrics, indicating that warmer conditions are associated with earlier start of season (SOS) and faster progression into summer. These relationships are relatively strong for American Elm ($r \approx -0.35$ to -0.40), suggesting a high sensitivity to temperature-driven early-season dynamics. Ginkgo exhibits similar but slightly weaker correlations ($r \approx -0.25$ to -0.30), indicating a more moderate response.

In contrast, late-season relationships are weaker and more variable. For American Elm, correlations between temperature and autumn or end-of-season timing remain negative but are smaller in magnitude ($r \approx -0.20$ to -0.30). Ginkgo shows even weaker and less consistent correlations in these stages (generally $r \approx -0.15$ to -0.25), reflecting greater variability in its phenological response. Precipitation effects are mixed for both species, with relatively small correlations across metrics (typically $|r| < 0.20$).

Overall, these patterns reinforce the importance of species-level differences in shaping phenological responses. While both species respond to climate drivers, American Elm appears more tightly coupled to temperature variation, whereas Ginkgo demonstrates a more variable and potentially resilient phenological response, particularly in later stages of the growing season.

Seasonal Differences in Predictability:

An important finding of this study is the variation in explanatory power across phenological stages. Models of spring and early-season dynamics (SOS and summer shoulder) exhibit higher explanatory power than those of autumn and late-season dynamics. This suggests that early-season phenology is more tightly linked to measurable environmental drivers, particularly temperature.

In contrast, late-season processes appear more variable and less predictable. The lower explanatory power of autumn models indicates that additional factors—such as photoperiod, cumulative stress, or localized management practices—may play a more important role. This pattern is consistent with broader phenological research, which finds that spring events are more strongly temperature-driven, while autumn dynamics are influenced by a wider range of interacting factors (Gao et al., 2023).

Implications and Limitations:

Taken together, these findings suggest that urban tree phenology is shaped by a combination of climate, urban form, and species composition. While temperature remains a dominant driver, urban conditions and biological differences introduce substantial variation that cannot be explained by climate alone. This has important implications for predicting phenological responses under future climate scenarios, as urban environments may not serve as simple proxies for broader warming trends.

Several limitations should be noted. First, climate variables are measured at the yearly level, which constrains the ability to fully separate their effects from broader temporal variation. Second, while impervious surface provides a high-resolution measure of urban intensity, it may not capture all relevant aspects of the urban environment, such as microclimate variation or management practices. Finally, the relatively low explanatory power of autumn models suggests that additional variables—such as photoperiod or cumulative stress—may be needed to better capture late-season dynamics.

Overall, this study provides strong evidence that phenological responses in urban environments are not solely driven by climate, but instead reflect a combination of environmental, spatial, and biological factors.

Conclusion

This study provides a comprehensive analysis of the drivers of tree phenology in Washington, DC using high-resolution, tree-level data. The findings show that phenological patterns are influenced by climate conditions, urban environmental factors, and species-level differences.

Spring temperature emerges as the most consistent and influential driver, with warmer conditions leading to earlier and more rapid seasonal development. However, urban conditions—particularly impervious surface—also play a significant role, especially in shaping late-season dynamics. In addition, substantial variation across species highlights the importance of biological factors in mediating environmental responses.

Importantly, these results demonstrate that urban phenology cannot be understood through climate alone. Instead, the interaction between environmental conditions, urban form, and species composition must be considered. This is particularly relevant for urban ecology, where species selection and landscape structure are shaped by human decisions and can influence how ecosystems respond to environmental change.

Future research should further explore late-season processes and incorporate additional variables, such as photoperiod and physiological stress, to better capture autumn dynamics. Expanding this analysis across additional cities and longer time periods would also improve understanding of how urban ecosystems respond to both climate change and urbanization.

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